

oxbotica

UNIVERSAL AUTONOMY

January 2021

We built an Autonomy Software Platform

It allows us to build off-road, industrial and on-road applications faster

It delivers recurring software revenue licenses

It offers performance and function unique to market

Built by best-in-class team

Key commercialisation partnerships established

Oxbotica Timeline

2014

Founded by Oxford University professors Paul Newman & Ingmar Posner

2015

Launch of Selenium and Caesium Products

2016

First company to test self-driving vehicle in public in the UK

Signed collaboration agreement with AXA XL

2017

Launch of Oxbotica software in Greenwich as part of the GATEway shuttle project

Partnership with Ocado to run an autonomous delivery service in London

Launch of the DRIVEN consortium

2018

Driverless shuttle trials at Gatwick Airport

Driverless trials airside at Heathrow Airport

Integration of a part of Selenium with OEM - trials run in Asia

Partnership with Addison Lee announced

Successful autonomous trial in a mine

2019

Deployment of a fleet of autonomous vehicles in London's challenging and complex urban environment

ZF collaboration announcement

2020

Commercial L4 deployment in mine

Announcement of collaborative radar-based project with Navtech

Integration & on-road deployment of our software with ZF hardware

Partnership announcement with Cisco to launch collaborative OpenRoaming platform

MoU signed with Wenco

First on-road MaaS trials across large UK cities

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Technology USP : Universal Autonomy Platform



ARCHITECTURE

- Modular
- Distributed
- Open API
- Designed for safety
- Robotics, optimization and ML substrates

USER NEEDS

FLEXIBILITY

- Sensor Agnostic
- Vehicle Agnostic
- Place Agnostic
- Industrial automation and on-road

MARKETS

EDGE

- Very Low Power
- CPU Agnostic
- Ease of Integration
- Universal Localisation
- Universal Perception

COST

CLOUD

- Fully web-based management solution
- Scheduling and event management
- Mapping and map management
- Simulation for Validation
- Full data synthesis and ML backbone

SCALING

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Technology USP : Universal Localisation



Surpasses all other solutions in both reliability and precision

Allowing any off-road markets to be addressed. Works where GPS does not.

Allows operation in conditions other solutions fail in (e.g. radar). Flexibility in cost

Addresses markets for all vehicle types - regardless of cost or shape

Removes dependency on third parties, allows verification and lowers cost. Allows operation anywhere. Zero installation

Full map provenance and management system. Independence of sensors addresses safety concerns

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Autonomy Platform Architecture. Individual solutions (components) are standalone for early revenue

USERS AND OPERATORS

AV Data and Fleet Management (called Caesium)

- Cloud-based data and vehicle management suite

Fleet monitoring and control

Data management and analytics

Cloud interface and security

In Cloud

Autonomy Platform (called Selenium)

- End-to-end autonomy platform
- Modular architecture enabling individual modules to be integrated into other AV software architectures

Foundation Autonomy

- Provides robust technical infrastructure to scale the solutions effectively

Localisation | Perception | Mapping | Tracking & Prediction | Planning & Control | Simulation

Foundation Autonomy

On Vehicle

VEHICLE

Technology USP : Universal Perception



Obstacle detection & identification , size and distance, velocity, free space & scene depth. With and without ML for safety and consensus.

Provides lower cost AI's. Allows rapid customization to specific use cases without needing to pay for training.

Allows operation where other solutions fail in (e.g. radar). Increases safety through consensus. Allows lower cost solutions

Fastest solution on market. Ultra low power, small footprint for commercialisation

Extending perception into the future, provides data fusion and estimates of future state, where and how will things move. Allows easy integration.

Full sensor suite, and safety architecture allows use in any domain making multiple markets addressable.

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Market Selection & Sequencing with Autonomy Platform



Autonomy Focus

- Current technology and go-to-market partnerships focus on Mining, Shuttles (People Carrier) and Goods Delivery
- Platform-level flexibility to add new industry solutions over time

Mining | Airports | Ports | Shuttles | Goods Delivery | Other..

Oxbotica Leadership Team



Ozgur Tohumcu, CEO

- Joined as CEO in 2019
- Previously CEO for Tantalum Corporation and held global executive leadership positions at Ericsson, Sapient & CGI
- Deep domain expertise within Automotive, Telecommunications and IoT
- MSc in Management and Technology from University of Pennsylvania



Marta Ostroumoff, Financial Director

- Joined as Finance Director in early 2018
- Previously Head of Finance for Prodrive, a UK based auto engineering group
- Also served as Cost Control Manager at Stork Sealing Systems and SE Controls
- BSc (Hons) in Business administration and management from Birmingham City University and Fellow Chartered Accountant



Ben Upcroft, VP of Technology

- Joined as Director of Projects in 2016 and is currently responsible for technology and development
- Previously Associate Professor in Robotics at QUT, focusing on applying computer vision to field robots Visiting Researcher at Oxford University together with Paul Newman



Tony Lowe, Interim COO

- Joined as Interim COO in July 2020
- Worked with Ozgur at Chief Delivery Officer at Tantalum before providing C-level consulting into various start ups and scale up companies
- Previously Co-founder of Jaguar Land Rover Connected Car department - led worldwide deployment
- Extensive operational, delivery and scaling domain experience



Paul Newman, Founder & CTO

- BP Professor of Information Engineering at University of Oxford and previously research Scientist at Massachusetts Institute of Technology
- Pioneer of solutions to the SLAM problem
- Serves on the UK Government's Department for Transport Science Advisory Council and the BSI Standards Committee for AV technology



Richard Jinks, VP of Commercial

- Joined as VP of Commercial in 2019 and is responsible for all commercial functions
- Previously held multiple leadership positions most recently at AMX SL as Global Head of Portfolio Solutions (and Global Autonomy Programme Lead
- Over 20 years experience of developing and leading Partnerships with large Corporate and Global partners



Graeme Smith, SVP of External Affairs

- Original Oxbotica CEO from 2014-2019, currently responsible for external affairs including grants, government interaction, IP Management and Safety
- More than 25 years' experience of automotive product/service development and program management
- 17+ years of experience at Ford Motor Company to Director level, responsible for creation of Ford's Telematics business and various Ford JV's
- PhD (Robotics), MBA, FIET, Chartered Engineer